# Chise Ishii

Graduate School of Science and Technology
Keio University, Japan

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## Research Interests

- Optimization, Convergence Analysis
- Dynamical Systems, ODEs / SDEs, Lyapunov Analysis
- Machine Learning, Deep Learning, Artificial Intelligence

## Education

**Ph.D., Engineering, Keio University, Japan** — April 2025 – Present
Graduate School of Science and Technology
Department of Open System Management

**M.S., Science, Waseda University, Japan** — April 2020 – March 2022
Graduate School of Fundamental Science and Engineering
Department of Pure and Applied Mathematics

**B.S., Science, Waseda University, Japan** — April 2016 – March 2020
School of Fundamental Science and Engineering
Department of Mathematics (Minor in Applied Mechanics and Aerospace Engineering)

UCLA Extension, USA — Summer 2017
American Language Center (ALC)

## Publications

1. Chise Ishii and Yasushi Narushima, "Convergence Analysis via ODE Approach for Convex Optimization with Linear Equality Constraints," *arXiv preprint*, May 2026. *arXiv: 2605.17819*

## Presentations

1. Chise Ishii and Yasushi Narushima, "Convergence Analysis via ODE Approach for Convex Optimization with Linear Constraints," The 24th Conference of the International Federation of Operational Research Societies (IFORS 2026), 2026. *Presentation type: Oral presentation*
2. Chise Ishii and Yasushi Narushima, "Convergence Analysis and Discretization Methods via an ODE Approach for Convex Optimization Problems with Linear Constraints," The 22nd Annual Meeting of the Japan Society for Industrial and Applied Mathematics, March 11, 2026. *Presentation type: Oral presentation*
3. Chise Ishii and Yasushi Narushima, "Convergence Analysis via an ODE Approach for Convex Optimization Problems with Equality Constraints," The 28th Information-Based Induction Sciences Workshop (IBIS 2025), November 13, 2025. *Presentation type: Poster presentation*
4. Chise Ishii and Yasushi Narushima, "Analysis of the Convergence Behavior of ADMM-Type Algorithms via an ODE Approach," RIMS Joint Research Workshop, "Mathematical Optimization: Frontiers of Mathematical Optimization," Kyoto University, September 1, 2025. *Presentation type: Oral presentation*

## Research Experience

Collaborative Researcher, Keio AI Research Center — April 2025 – Present
Kanagawa, Japan

## Industry Experience

Banker, Mitsubishi UFJ Financial Group, Inc. and MUFG Bank, Ltd. — April 2022 – Present
Tokyo, Japan

Machine Learning Engineer, AIREV Inc. — Dec 2019 – Jan 2022
Tokyo, Japan

Software Development (Student Intern), Fujitsu — Summer 2020
Tokyo, Japan

## Teaching Experience

Applied Practical Data Science and Artificial Intelligence 2B — June 2026
Lecturer, Tokyo University of Science

Programming for Computer Algebra — Fall 2020 / 2021
Teaching Assistant, Waseda University

Introduction to Computing — Spring 2020 / 2021
Teaching Assistant, Waseda University

Mathematics B2 (Calculus) — 2019
Teaching Assistant, Waseda University

## Academic Service

- Reviewer: American Control Conference

## Fellowships

- "Establishing Fundamental Theory for Robust and High-Performance Deep Learning and Machine Learning Methods and Their Applications," Doctoral Student Research Support Program, Ushioda Memorial Foundation, April 2026 – March 2027.